Diskretne Coonsove ploskve

Rok Filipovič, Jernej Jezeršek

15. januar 2025

Pregled

1. Coonsove ploskve
2. Diskretne Coonsove ploskve
3. Trajnostne ploskve
4. Primeri
5. Optimizacija

Coonsove ploskve

(Zvezna) Coonsova ploskev

Ploskev, ki interpolira robne krivulje $\mathbf{x}(0,v)$, $\mathbf{x}(1,v)$, $\mathbf{x}(u,0)$, $\mathbf{x}(u,1)$:

$$\begin{aligned}\mathbf{x}(u,v) &= (1-u)\mathbf{x}(0,v) + u\mathbf{x}(1,v) \\ &\quad + (1-v)\mathbf{x}(u,0) + v\mathbf{x}(u,1) \\ &\quad - \begin{bmatrix} 1-u & u \end{bmatrix} \begin{bmatrix} \mathbf{x}(0,0) & \mathbf{x}(0,1) \\ \mathbf{x}(1,0) & \mathbf{x}(1,1) \end{bmatrix} \begin{bmatrix} 1-v \\ v \end{bmatrix}\end{aligned}$$

Temeljni načeli

Načelo minimuma

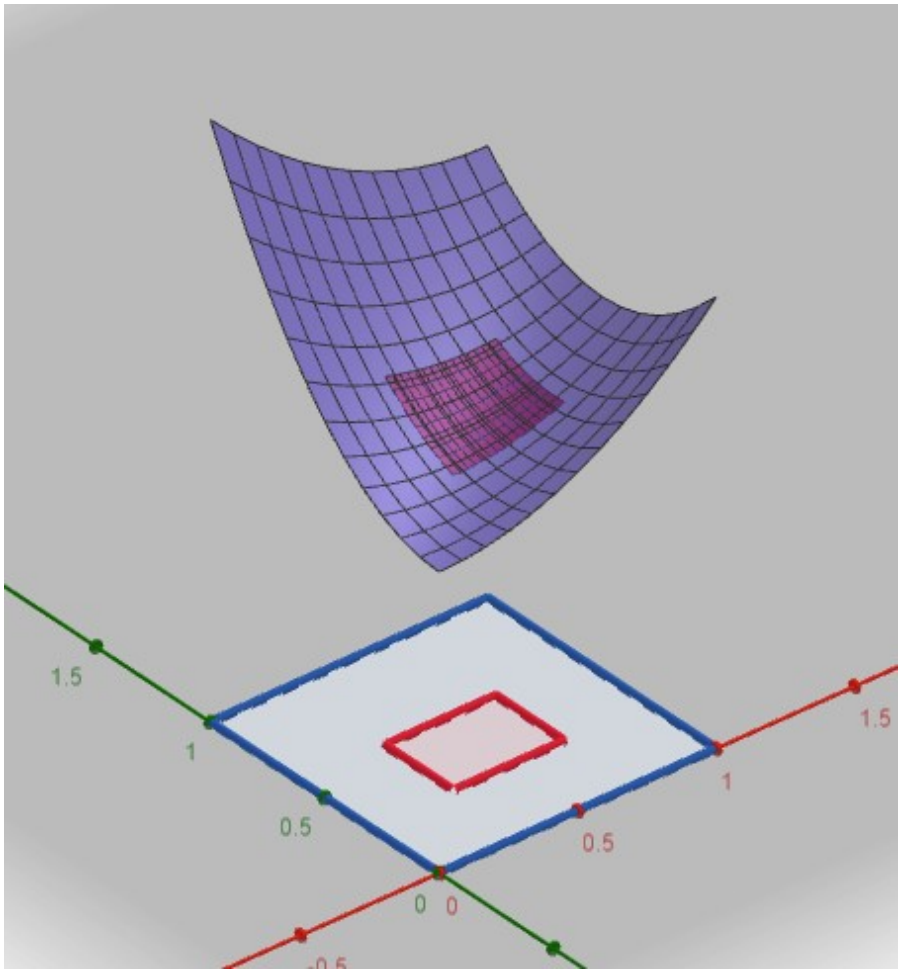
$$min\left(\int_U \mathbf{x}_{uv}^2 \, dS\right)$$

Euler-Lagrangeeva PDE: $\mathbf{x}_{uuvv} = 0$

Temeljni načeli

Načelo trajnosti

Coonsova ploskev podogrodja = originalna Coonsova ploskev zožana na podogodje



Diskretne Coonsove ploskve

$$\begin{matrix} \mathbf{b}_{00} & \mathbf{b}_{01} & \dots & \mathbf{b}_{0n} \\ \mathbf{b}_{10} & & & \mathbf{b}_{1n} \\ \vdots & & & \vdots \\ \mathbf{b}_{m0} & \mathbf{b}_{m1} & \dots & \mathbf{b}_{mn} \end{matrix}$$

Diskretna Coonsova ploskev

Notranje točke, ki ustrezajo diskretnemu ogrodju

$$\begin{aligned} \mathbf{b}_{i,j} = {} & (1 - i/m)\mathbf{b}_{0,j} + \frac{i}{m}\mathbf{b}_{m,j} \\ & + (1 - j/n)\mathbf{b}_{i,0} + \frac{j}{n}\mathbf{b}_{i,n} \\ & - \begin{bmatrix} 1 - i/m & i/m \end{bmatrix} \begin{bmatrix} \mathbf{b}_{0,0} & \mathbf{b}_{0,n} \\ \mathbf{b}_{m,0} & \mathbf{b}_{m,n} \end{bmatrix} \begin{bmatrix} 1 - j/n \\ j/n \end{bmatrix} \end{aligned}$$

Temeljni načeli

Načelo minimuma

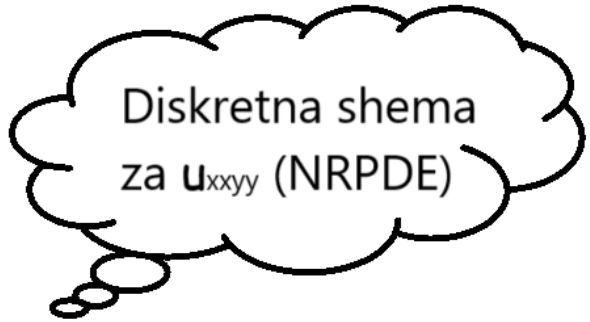
$$min \left(\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} (\Delta^{1,1} \mathbf{b}_{i,j})^2 \right)$$

Načelo trajnosti

Zožitev na 3x3:

$$\mathbf{b}_{i,j} = -\frac{1}{4} \left(\mathbf{b}_{i-1,j+1} + \mathbf{b}_{i+1,j+1} + \mathbf{b}_{i-1,j-1} + \mathbf{b}_{i+1,j-1} \right) + \frac{1}{2} \left(\mathbf{b}_{i,j+1} + \mathbf{b}_{i-1,j} + \mathbf{b}_{i+1,j} + \mathbf{b}_{i,j-1} \right)$$

Diskretna Euler-Lagrangeeva PDE



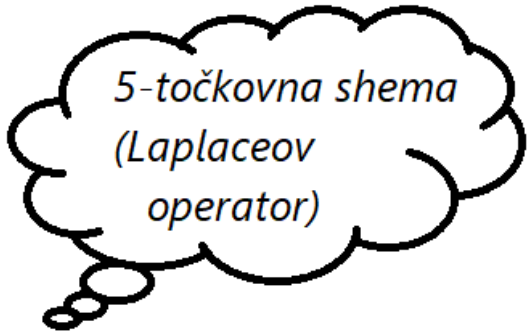
$$\mathbf{b}_{i,j} = \frac{1}{4} \times \begin{matrix} -1 & 2 & -1 \\ 2 & \bullet & 2 \\ -1 & 2 & -1 \end{matrix}$$

Trajnostne ploskve

$$\mathbf{b}_{i,j} = \begin{matrix} \alpha & \beta & \alpha \\ \beta & \bullet & \beta \\ \alpha & \beta & \alpha \end{matrix}$$

- $\alpha + \beta = 1/4$
- DCP: $\alpha = -1/4$
- Različne oblike

Diskretni Laplaceov operator



$$\mathbf{b}_{i,j} = \frac{1}{4} \times \begin{matrix} 0 & 1 & 0 \\ 1 & \bullet & 1 \\ 0 & 1 & 0 \end{matrix}$$

Končna maska

$$\mathbf{b}_{i,j} = \frac{1}{4}\left(e \times \begin{matrix} -1 & 2 & -1 \\ 2 & \bullet & 2 \\ -1 & 2 & -1 \end{matrix} + (1-e) \times \begin{matrix} 0 & 1 & 0 \\ 1 & \bullet & 1 \\ 0 & 1 & 0 \end{matrix} \right) \qquad \begin{matrix} e \in [0,1] \\ (\alpha = -e/4) \\ \alpha \in [-1/4, 0] \end{matrix}$$

"... no geometry-independent combination of Euler–Lagrange and Laplace masks will be sufficient for all geometries."

Računanje trajnostne ploskve

- Za mrežo velikosti $(m+1) \times (n+1)$ imamo $(m-1)(n-1)$ neznank.
- Za vsako neznanko skonstruiramo linearno enačbo

$b_{0,0}$ α	$b_{0,1}$ β	$b_{0,2}$ α	$b_{0,3}$	$b_{0,4}$
$b_{1,0}$ β	$b_{1,1}$ •	$b_{1,2}$ β	$b_{1,3}$	$b_{1,4}$
$b_{2,0}$ α	$b_{2,1}$ β	$b_{2,2}$ α	$b_{2,3}$	$b_{2,4}$
$b_{3,0}$	$b_{3,1}$	$b_{3,2}$	$b_{3,3}$	$b_{3,4}$
$b_{4,0}$	$b_{4,1}$	$b_{4,2}$	$b_{4,3}$	$b_{4,4}$

$$\begin{aligned}\alpha b_{0,0} + \beta b_{0,1} + \alpha b_{0,2} \\ +\beta b_{1,0} - b_{1,1} + \beta b_{0,2} \\ +\alpha b_{2,0} + \beta b_{2,1} + \alpha b_{2,2} = 0\end{aligned}$$

$$\begin{aligned}- b_{1,1} + \beta b_{0,2} + \beta b_{2,1} + \alpha b_{2,2} = \\ -\alpha b_{0,0} - \beta b_{0,1} - \alpha b_{0,2} - \beta b_{1,0} - \alpha b_{2,0}\end{aligned}$$

Računanje trajnostne ploskve

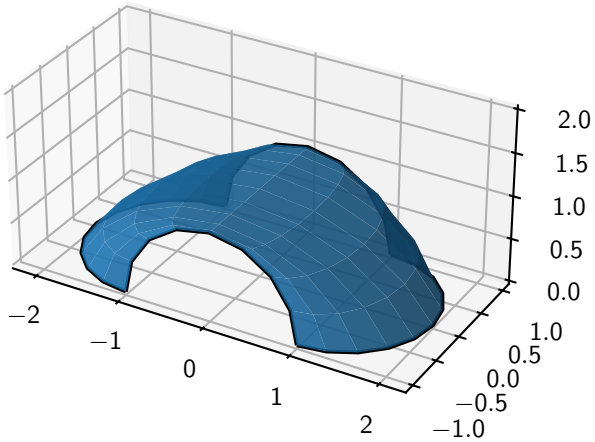
$$0 = \sum_{x=0}^{2}\sum_{y=0}^{2} b_{i-y-1,j-x-1}k_{y,x} \;\; \forall i = 1,\ldots,m-1 \;\; \forall j = 1,\ldots,n-1$$

$$\begin{bmatrix} k_{0,0} & k_{0,1} & k_{0,2} \\ k_{1,0} & k_{1,1} & k_{1,2} \\ k_{2,0} & k_{2,1} & k_{2,2} \end{bmatrix} = \begin{bmatrix} \alpha & \beta & \alpha \\ \beta & -1 & \beta \\ \alpha & \beta & \alpha \end{bmatrix}$$

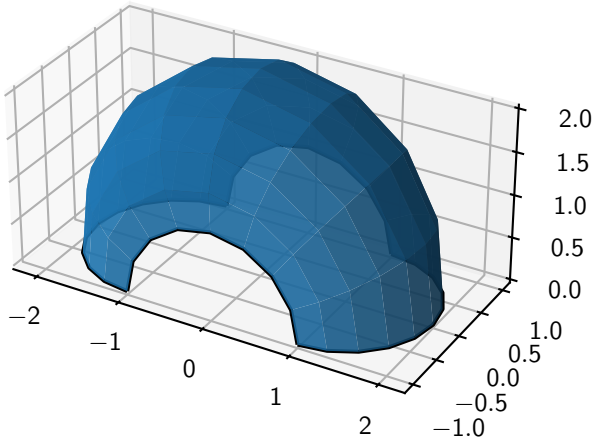
- Linearne enačbe shranimo v matrični obliki $Ax = b$
- V trodimenzionalnem primeru so $b_{i,j}$ vektorji in moramo izračunati vsako komponento posebej.

Primeri

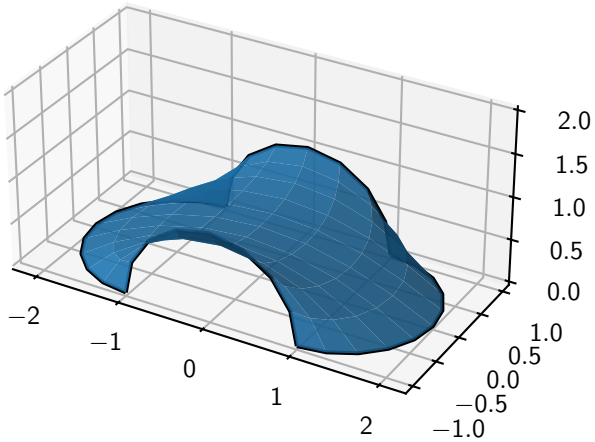
$\alpha = -0.25$



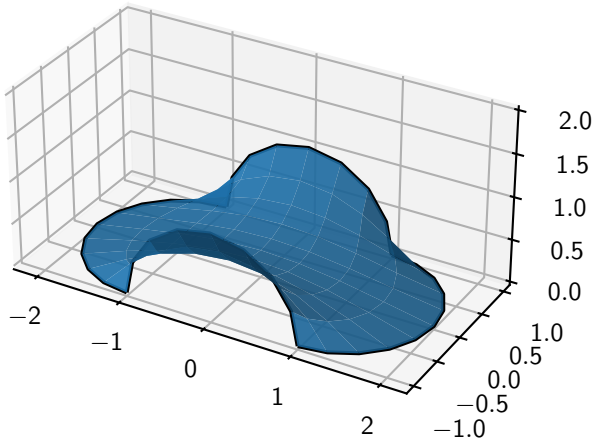
$\alpha = -0.26$



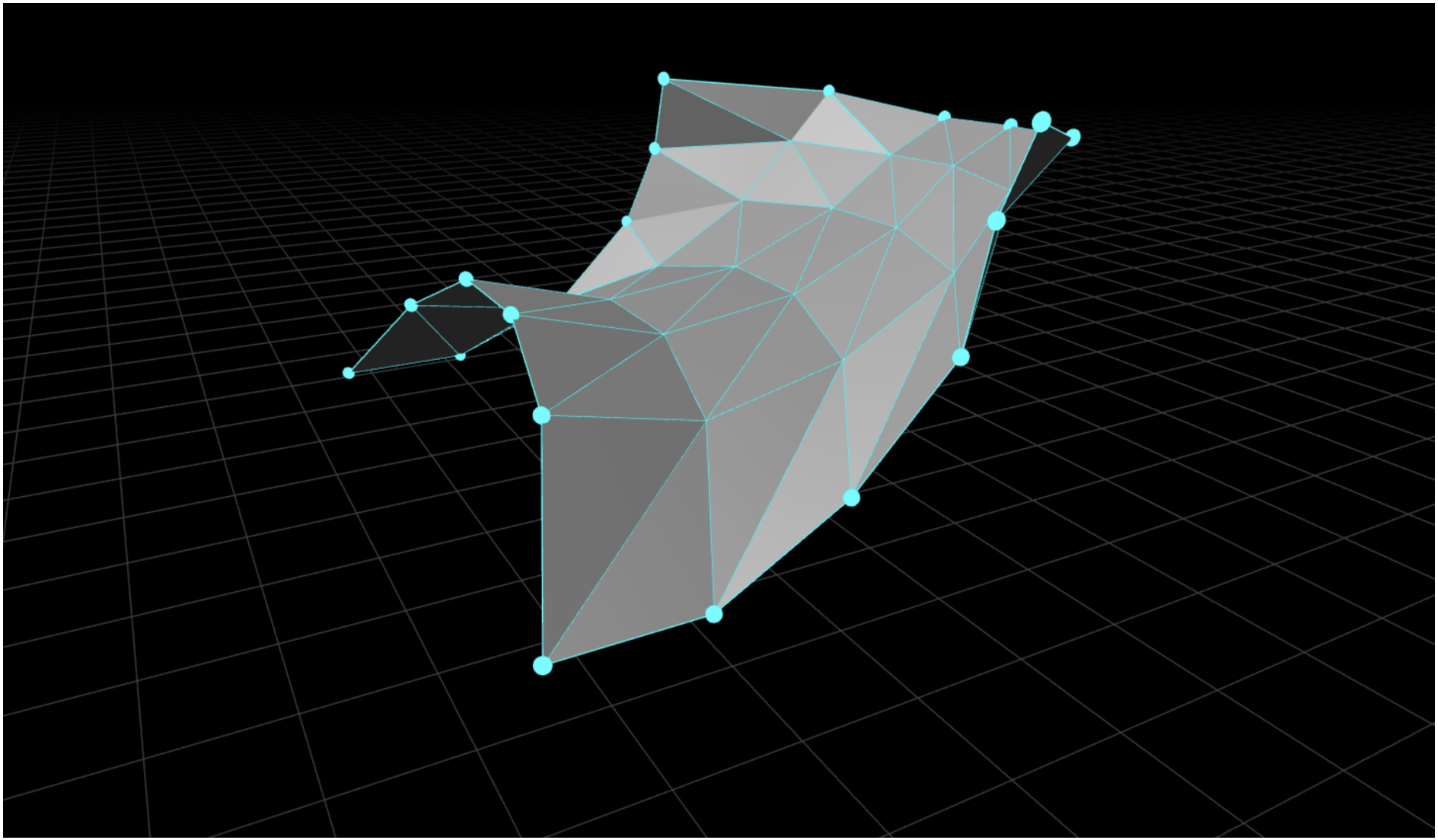
$\alpha = -0.23$



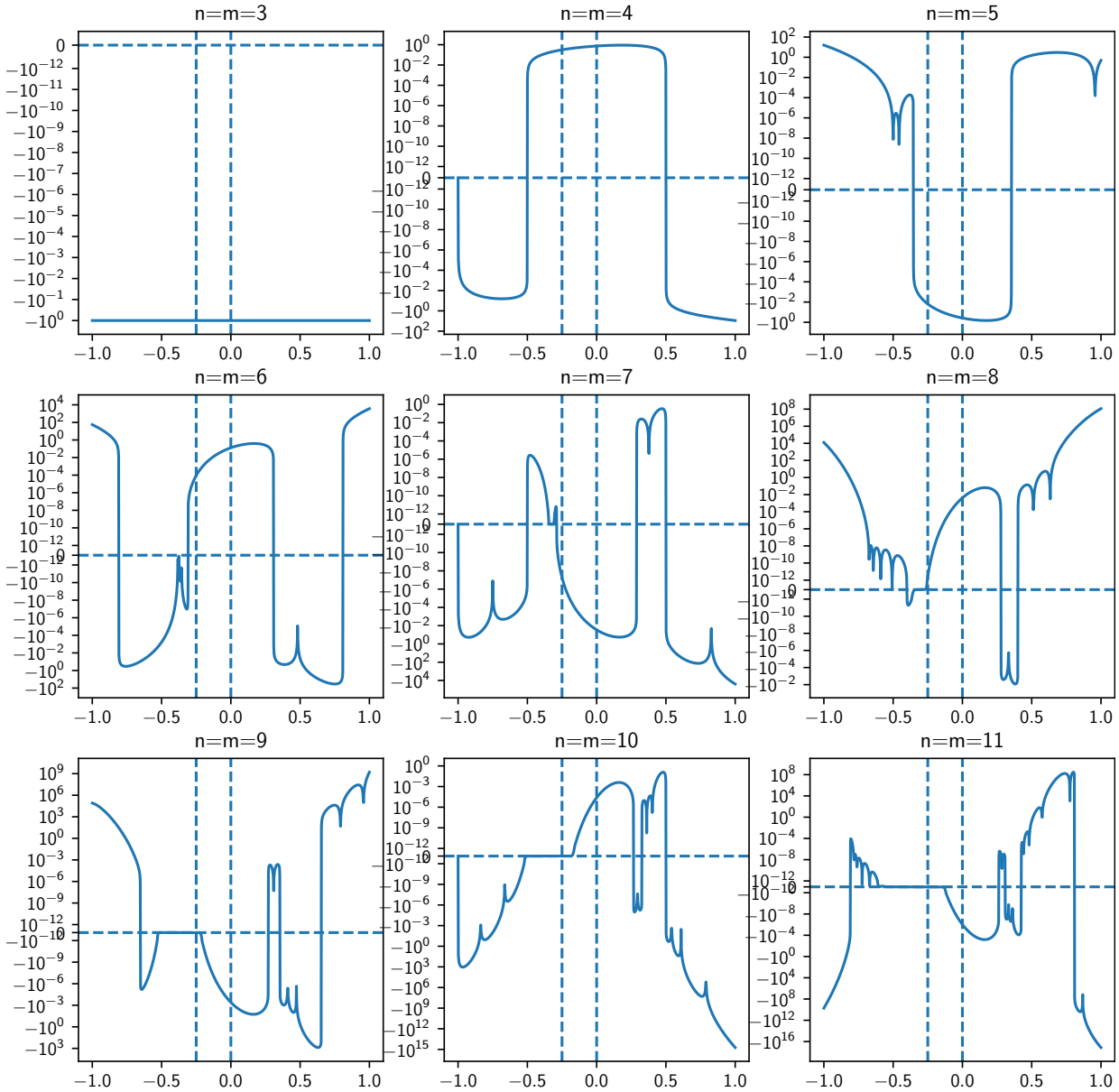
$\alpha = 0$



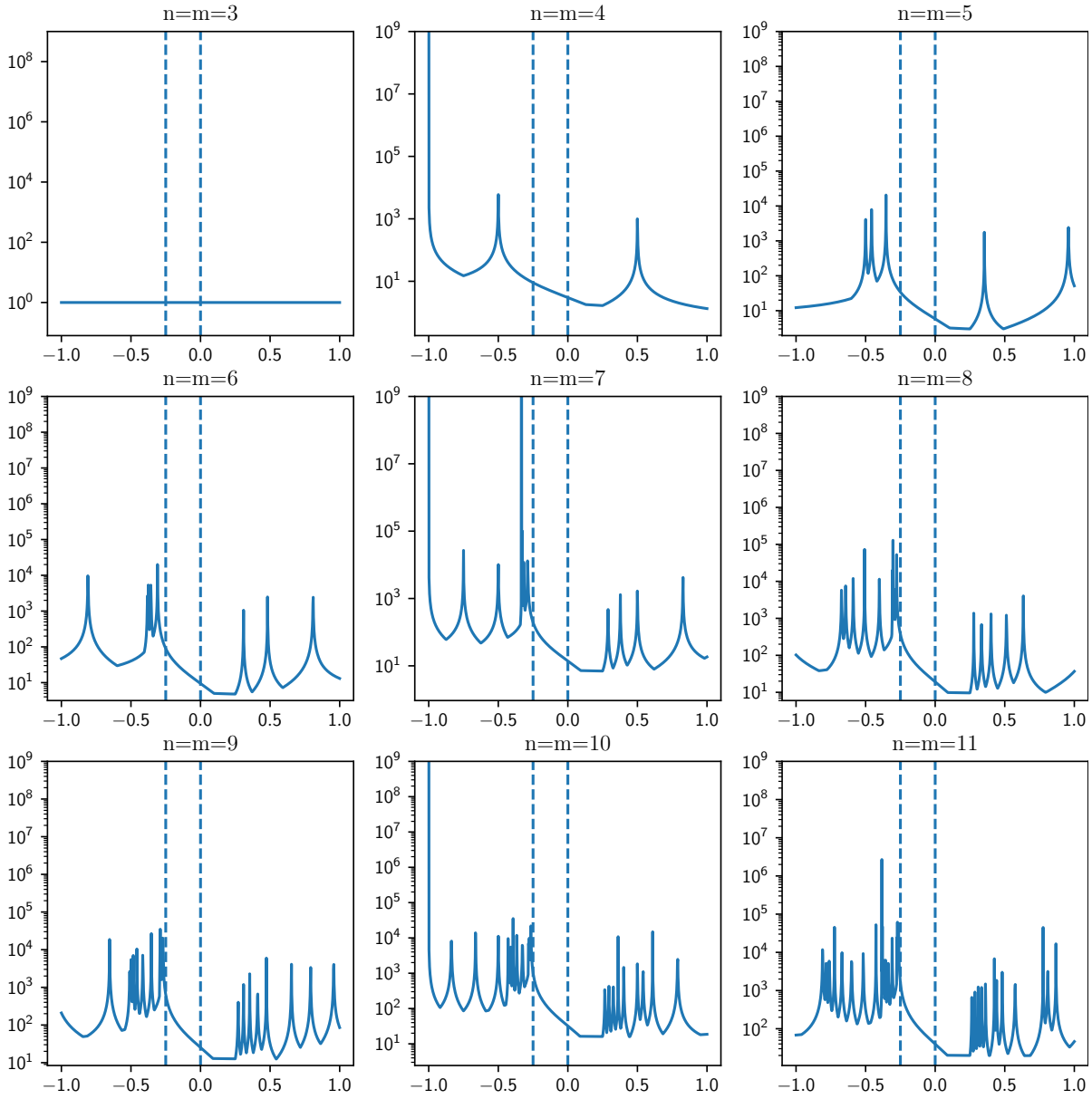
Interaktivna vizualizacija



Determinanta matrike A



Občutljivost matrike A



Iskanje α

- Izbira α močno vpliva na obliko ploskve
- Ne obstaja ena sama vrednost α, ki bi dala dobre rezltate za vse izbire točk.

"Good here refers to the traditional designer's paradigm that the interior of a surface should not have different shape characteristics than those implied by the boundary curves."

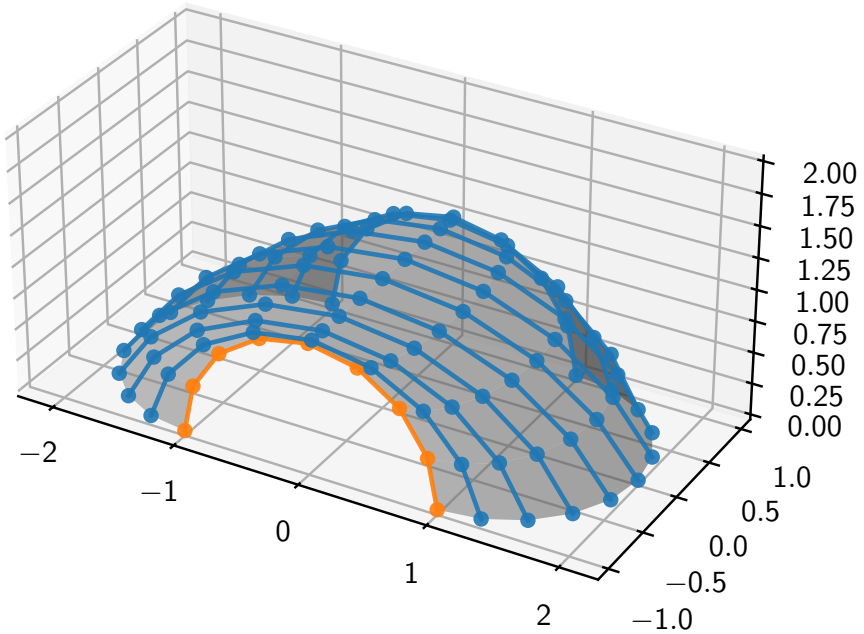
Matematična definicija dobre ploskve

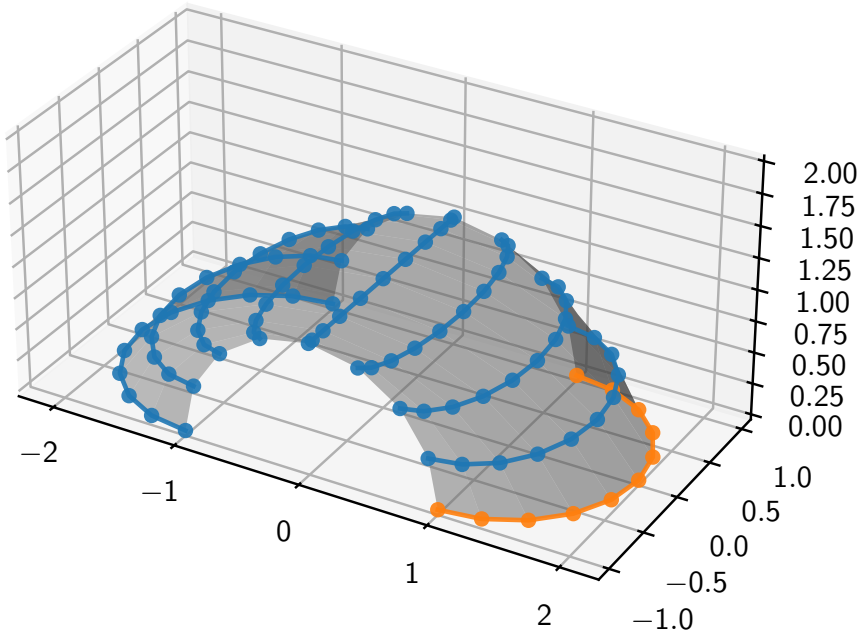
- Definiramo funkcijo, ki nam pove v kakšni meri oblika ploskve odstopa od oblike, ki jo implicirajo robne krivulje
- Vsako robno krivuljo primerjamo z vzporenimi notranjimi krivuljami
- Seštevamo razlike v kotih

$$s_{i,j} = \frac{b_{i+1,j} - b_{i,j}}{\|b_{i+1,j} - b_{i,j}\|} \quad \forall i = 0, \ldots, m-1 \; \forall j = 0, \ldots, n$$

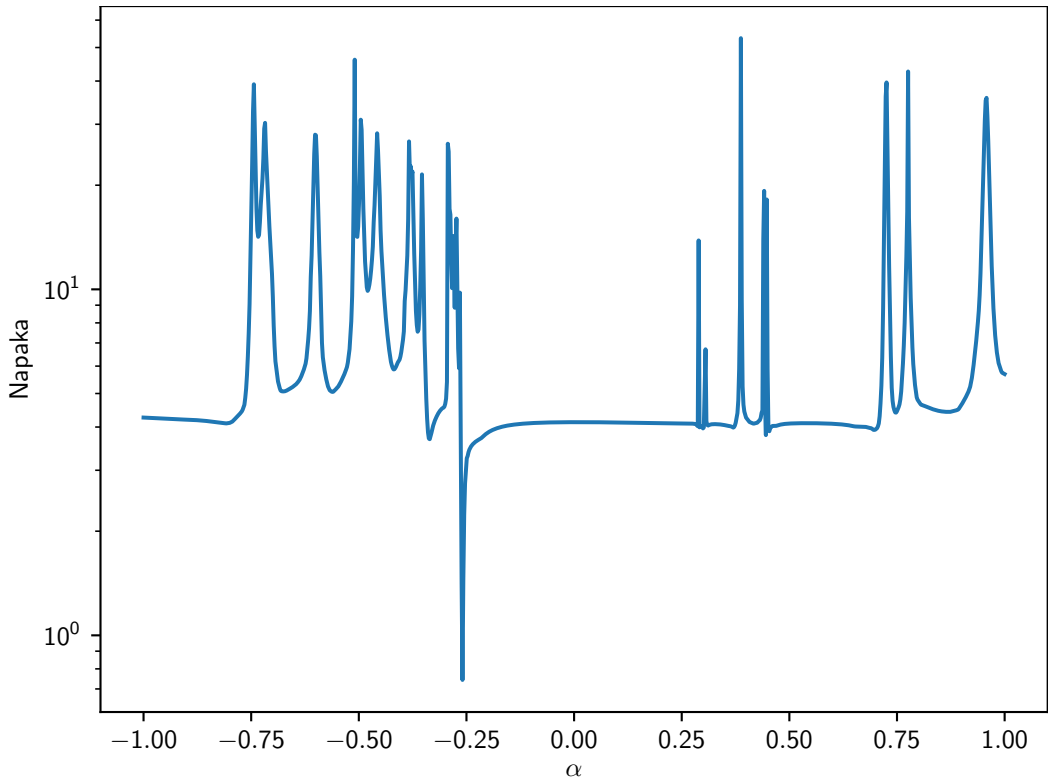
$$\varphi_{i,j} = \arccos\left(\langle s_{i,j}, s_{i+1,j} \rangle\right)$$

Matematična definicija dobre ploskve



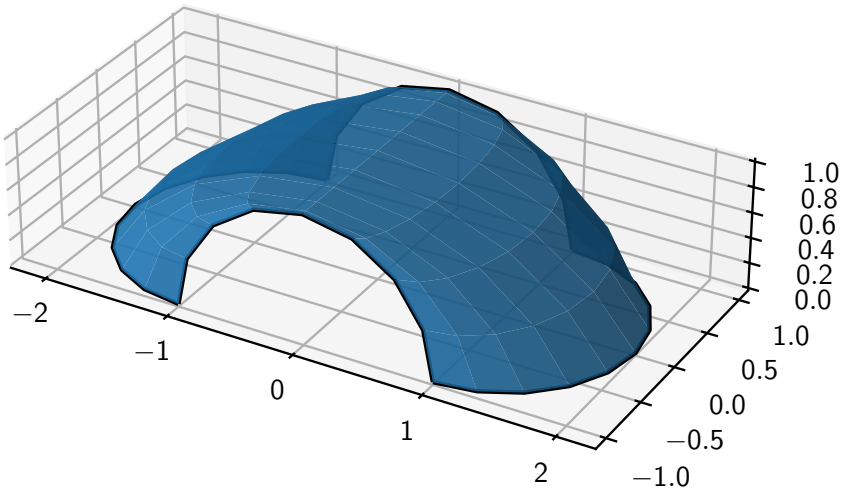


Primeri

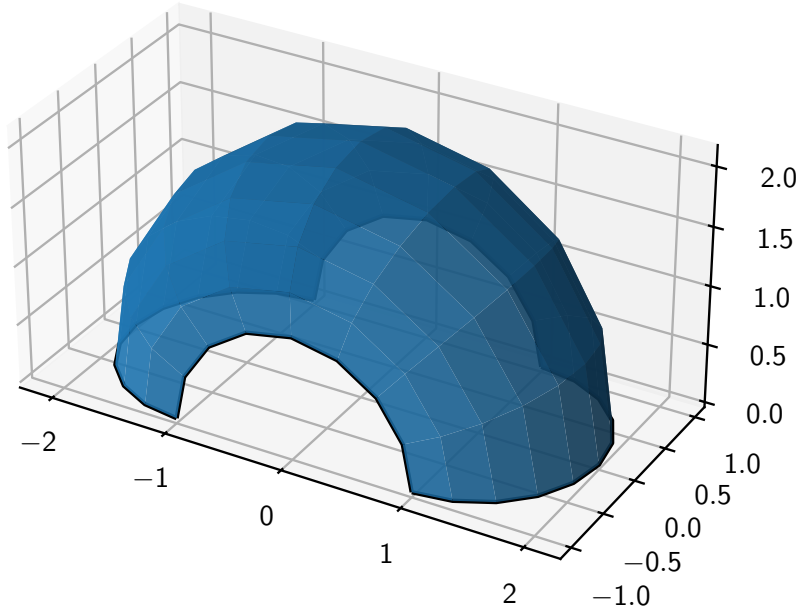


Primeri

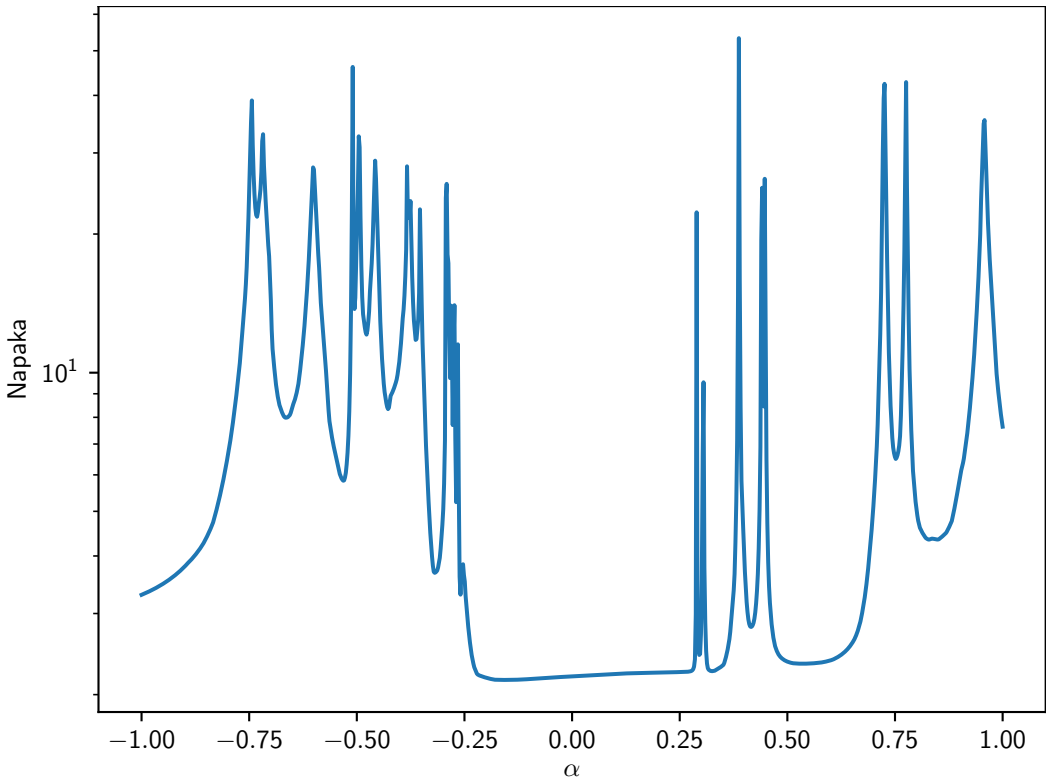
$\alpha = -0.25$ (*Coons*)



$\alpha = -0.2593$ (*optimizirana*)

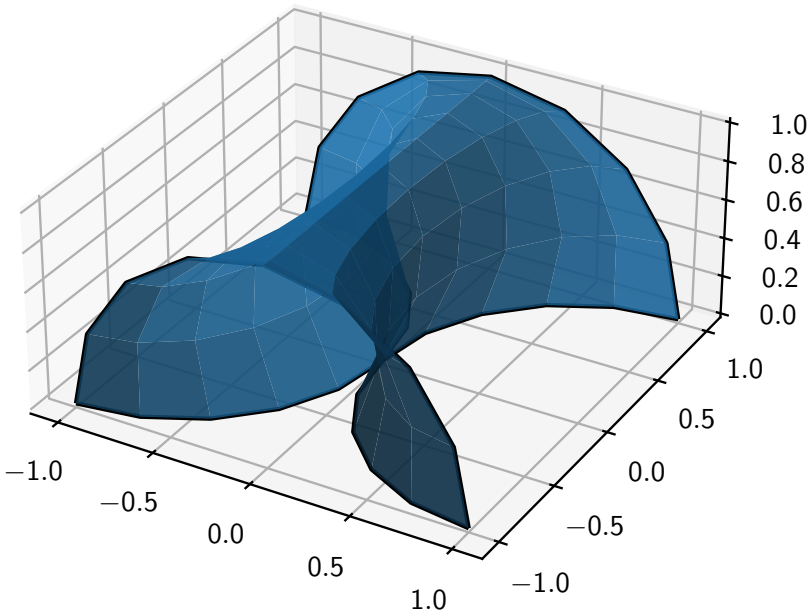


Primeri

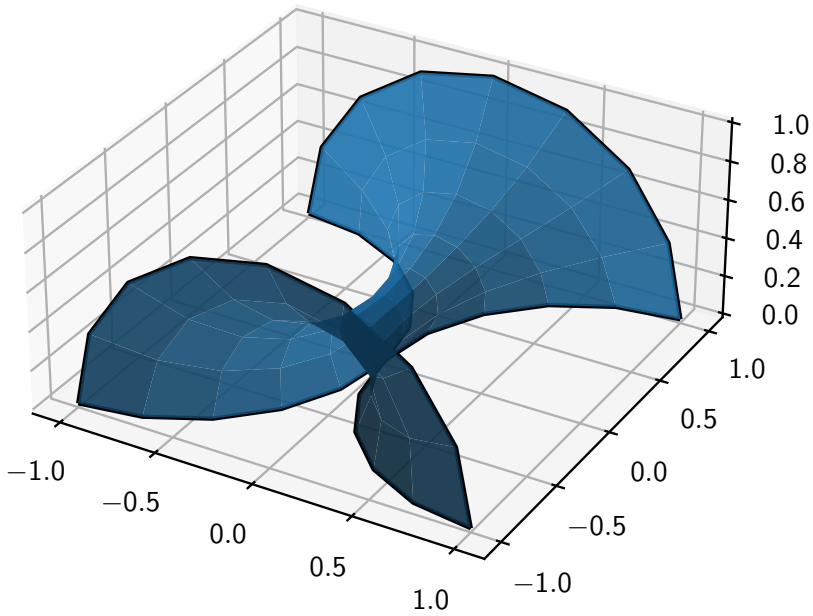


Primeri

$\alpha = -0.25$ (*Coons*)



$\alpha = -0.1572$ (*optimizirana*)



Viri

- G. Farin, D. Hansford: Discrete Coons patches, Computer Aided Geometric Design 16, 1999, 691–700.